THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of isosceles triangles with whole-number sides and perimeter 14 cm (equilateral included) is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

2. $\frac{5}{6} \div (\frac{5}{3} \div \frac{2}{3})$ equals: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{3}$

3. Two supplementary angles are in the ratio $2 : 3$. The larger angle is: 1

(a) 128° **(b)** 98° **(c)** 118° **(d)** 108°

4. The smallest 4-digit number divisible by 9 is: 1

(a) 1017 **(b)** 1007 **(c)** 1009 **(d)** 1008

5. The number of triangles with whole-number sides (in cm) and perimeter 12 cm is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

6. $\frac{3}{7}$ of a number is 12. Three times the number is: 1

(a) $84\frac{1}{2}$ **(b)** 42 **(c)** 84 **(d)** 168

7. The supplement of the complement of 35° is: 1

(a) 135° **(b)** 125° **(c)** 145° **(d)** 115°

8. The largest 4-digit number with all digits different that is divisible by 5 is: 1

(a) 9876 **(b)** 9874 **(c)** 9877 **(d)** 9875

9. The exterior angle at the base of an isosceles triangle is 125°. The apex angle is: 1

(a) 70° **(b)** 60° **(c)** 90° **(d)** 80°

10. A pipe fills $\frac{3}{4}$ of a tank in 15 minutes. The time to fill the whole tank (minutes) is: 1

(a) 10 **(b)** 40 **(c)** $20\frac{1}{2}$ **(d)** 20

11. The reflex angle that goes with an angle of 135° (so that together they make a complete turn) is: 1

(a) 245° **(b)** 235° **(c)** 225° **(d)** 215°

12. Kaprekar's routine on 6321 reaches 6174 after how many steps? 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

13. The sum of the three exterior angles of a triangle (one at each vertex) is: 1

(a) 360° **(b)** 270° **(c)** 540° **(d)** 180°

14. $\frac{2}{5}$ of a wall is already painted. Then $\frac{2}{3}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{1}{10}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{11}{30}$

15. Two complementary angles differ by 30°. The larger angle is: 1

(a) 80° **(b)** 50° **(c)** 70° **(d)** 60°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. List all isosceles triangles with whole-number sides and perimeter 20 cm (equal sides a, base b). How many are there? 2

17. Find the fraction exactly halfway between $\frac{1}{3}$ and $\frac{1}{2}$, and verify that its distance from each is the same. 2

18. Find the angle whose supplement is 4 times its complement. 2

19. A two-digit number added to its reverse gives 143 and subtracted from it gives 45. Find the number. 2

20. In $\triangle ABC$ and $\triangle PQR$ it is known that $\angle A = \angle P = 40^\circ$, $\angle B = \angle Q = 65^\circ$ and $\angle C = \angle R = 75^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In $\triangle ABC$, $\angle A = 40^\circ$. The exterior angles at B and C are $(5x + 50)^\circ$ and $(4x + 62)^\circ$. Find x and the angles B and C. 3

22. Ravi alone can finish a job in 10 days and Sita alone in 15 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

23. Two supplementary angles are in the ratio $1 : 8$. Find the complement of the smaller angle and the supplement of the larger angle. 3

24. Start with 1259 and apply Kaprekar's routine until 6174 appears. Show every step. 3

OR

The perimeter of an isosceles triangle is 45 cm. Each equal side is 3 cm longer than the base. Find the sides and show that such a triangle exists. 3

25. A cyclist rides at $\frac{21}{2}$ km per hour for 4 hours, then at $\frac{35}{4}$ km per hour for 2 hours. Find the total distance and the average speed for the whole journey. 3

OR

Two adjacent angles form a linear pair and one of them is 70°. Find the other angle, the half of each angle and the angle between the two bisectors. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Three angles on one side of a straight line at a point are in the ratio $2 : 3 : 4$. Find the three angles and name the kind of each. 5

OR

Use reverse-and-add on 155 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

27. The exterior angles of a triangle (one at each vertex) are in the ratio $4 : 5 : 6$. Find the three exterior angles, the three interior angles and classify the triangle. 5

OR

A farmer's land is 360 acres. He sells $\frac{2}{5}$ of it, gives $\frac{1}{6}$ of the remainder to his son and plants $\frac{4}{5}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND